FPGA-BASED CUSTOM SMART CONTROLLER

Intermediate Verilog HDL Project | Xilinx Vivado | Simulation-only

Objective. Build a configurable sequential controller using an FSM, synchronizers, clock enable, countdown timer and configuration ROM.

1. Functional overview

The design models a generic appliance/process controller. It accepts start, pause, resume, stop, emergency, over-temperature, sensor-ready and fault-clear inputs. It drives motor and heater enables, an alarm, state debug and remaining-time outputs. No FPGA board is assumed.

2. Architecture

Asynchronous inputs pass through two-flop synchronizers. A clock-enable generator produces a one-cycle tick. A constant configuration ROM selects a duration of 5, 10, 20 or 30 seconds. The FSM controls the countdown timer and safe outputs. All sequential logic uses one system clock; tick is an enable, not a separate clock.

3. FSM

State	Function	Motor	Heater	Alarm
IDLE	Wait	0	0	0
STARTUP	Wait for ready	0	0	0
RUNNING	Active process	1	1	0
PAUSED	Hold timer	0	0	0
COMPLETE	Timer expired	0	0	0
FAULT	Fault latched	0	0	1

FAULT has priority over normal operation and remains latched until clear_fault is asserted while emergency and over-temperature inputs are inactive.

4. Verification

The testbench checks reset, start, duration loading, pause/count hold, resume, completion, emergency fault, fault clearing and over-temperature fault. The expected successful console message is: PASS: all smart controller tests completed with zero errors. Run the testbench in Vivado and capture genuine waveforms and console output.

5. Synthesis and timing

Run vivado/run_synth.tcl to generate utilization and timing-summary reports. The script uses xc7a35tcpg236-1 as an example part; change it if a real target is selected. Hardware implementation requires a valid clock constraint and board-specific XDC pin assignments. Since this project is simulation-only, no pin map is provided and no timing-closure claim is made.

6. Power and performance

The design uses a single clock domain and clock-enable pulses, reducing clocking complexity. Two-flop synchronizers reduce metastability propagation risk but do not debounce mechanical switches. Estimate power using realistic activity data and optimize only after reviewing implementation reports.

7. Project files

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rtl/: synchronizer, clock enable, ROM, timer, FSM, top module\ntestbench/tb_smart_controller.v\nvivado/run_sim.tcl, run_
```

8. Conclusion

This project demonstrates intermediate FPGA design concepts including sequential logic, FSM control, registers, a timer, configuration memory, simulation, synthesis and timing-report generation. Actual reports

and screenshots must be generated by running Vivado in the user's environment.